Tahmid Hasan Sakib

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RESEARCH INTERESTS

Autonomous systems; embedded AI and computer vision; adversarial perception; camera-LiDAR sensing; secure ROS 2 and cyber-physical systems; runtime monitoring and physical validation.

EDUCATION

Tennessee Technological University, Cookeville, Tennessee

Master of Science in Electrical and Computer Engineering, expected December 2026 | **GPA 4.00/4.00**

Master's thesis successfully defended; official degree conferral pending.

Thesis: "Securing Autonomous Vehicles Against Navigational Perception and Middleware Attacks: Experimental Validation on a 1/10-Scale Quanser Testbed."

Military Institute of Science and Technology, Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering, August 2021

RESEARCH EXPERIENCE

Graduate Research Assistant | Tennessee Technological University | May 2024-Present

- Validated camera, LiDAR, detector, and control workflows in collaborative QCar1 and QCar2 research, documenting platform-specific configurations for physical tests.
- Analyzed clean and adversarial vehicle runs to connect detector evidence and range estimates with stopping behavior.
- Characterized SROS2 provisioning and middleware trust boundaries on QCar2 and reported qualitative effects of authenticated-message manipulation.
- Compared offline and live evidence for runtime anomaly monitoring and image restoration with collaborators, recording failure cases and control outcomes.

JOURNAL ARTICLE ACCEPTED FOR PUBLICATION

Tahmid Hasan Sakib, Abhijeet Solanki, Wesam Al Amiri, Syed Rafay Hasan, Syed Ali Asad Rizvi, and Terry N. Guo. "Vulnerability Assessment of Adversarial Image Attacks in a Miniaturized Autonomous Vehicle Testbed without Relying on Way-Point Information." *IEEE Access*, accepted for publication, 2026.

PUBLISHED PEER-REVIEWED CONFERENCE PAPERS

1. Brianna D'Urso, **Tahmid Hasan Sakib**, Syed Rafay Hasan, and Terry N. Guo. "Physical Evaluation of Naturalistic Adversarial Patches for Camera-Based Traffic-Sign Detection." *2026 IEEE 2nd International Conference on Secure IoT, Assured and Trusted Computing (SATC)*, 2026. doi:10.1109/SATC69565.2026.11542547.
2. **Tahmid Hasan Sakib**, Yago Romano Martinez, Carter Brady, Syed Rafay Hasan, and Terry N. Guo. "Supply Chain Exploitation of Secure ROS 2 Systems: A Proof-of-Concept on Autonomous Platform Compromise via Keystore Exfiltration." *2025 IEEE Military Communications Conference (MILCOM)*, workshop paper, 2025. doi:10.1109/MILCOM64451.2025.11309897.
3. **Tahmid Hasan Sakib**, Md. Ishak, Farzana Faruk Jhumu, and Md. Ameer Ali. "Analysis of Suicidal Tweets from Twitter Using Ensemble Machine Learning Methods." *2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI)*, 2021. doi:10.1109/ACMI53878.2021.9528252.

CONFERENCE EXTENDED ABSTRACT AND POSTER

Tahmid Hasan Sakib, Wesam Al Amiri, Abhijeet Solanki, Syed Rafay Hasan, Terry N. Guo, and Syed Ali Asad Rizvi. "Investigating Adversarial Image Attacks in a Sensor Fusion Framework Using a Scaled Autonomous Vehicle Testbed." IFAC Modeling, Estimation and Control Conference (MECC), 2025. Extended abstract and poster contribution. doi:10.5281/zenodo.17497276.

MANUSCRIPT UNDER REVIEW

Tahmid Hasan Sakib, Abhijeet Solanki, Syed Rafay Hasan, Terry Guo, and Syed Ali Asad Rizvi. "PULSE: Real-Time Flagging of Patch-Induced Stop-Sign Misses in Autonomous Vehicles." Submitted and under review, IEEE Vehicular Technology Conference (VTC 2027-Spring).

MANUSCRIPT IN PREPARATION

Abhijeet Solanki, **Tahmid Hasan Sakib**, Syed Rafay Hasan, Terry Guo, and Syed Ali Asad Rizvi. "REDAP: Real-Time Edge Defense Against Patch Attacks in Autonomous Driving." CVPR 2027 registered; full paper submission pending.

SELECTED RESEARCH PROJECTS

QCar1 physical adversarial perception

A non-ROS Jetson TX2 platform combined CSI-camera YOLOv5n detection with RPLiDAR A2 ranging for three-stop navigation without sign waypoints. Across 30 clean runs and separate groups of 18 pre- and 18 post-mitigation printed-sign runs, runs with a missed stop changed from 8/18 to 0/18. Detector and runtime settings both changed.

QCar2 secure ROS 2 supply-chain evaluation

A controlled SROS2 proof of concept traced credential-provisioning and authenticated DDS trust boundaries. The physical evidence documents qualitative vehicle behavior effects, including sustained turning.

PULSE runtime monitoring and fallback

Clean-run-calibrated detector evidence flagged 18 of 20 patched-sign encounters in 10 live black/white QCar2 fallback-on runs; all 20 stop sequences completed. Recorded-log replay was a separate offline phase; stop-line position was not independently measured.

REDAP image restoration

On 151 stored QCar2 frames from two physical runs processed offline, 57/151 attacked and 110/151 restored frames met the 0.60 confidence cutoff. Later E3/E4 closed-loop demonstrations were separate and had mixed outcomes.

INDEPENDENT AND EARLIER ENGINEERING PROJECTS

Network Intrusion Detection for IoT Devices | Repository

Independent post-undergraduate project using CICIoT2023 for binary, 8-class, and 34-class classification. Committed Python notebooks cover preprocessing, feature selection, conventional ML, and ANN/LSTM/GRU experiments.

ANGTHAS telemedicine application

Undergraduate-era Node.js, Express, and MongoDB application with patient/doctor routes and modules for symptom checking, appointments, messaging, and video interaction.

Metro Rail Management System | Repository

Undergraduate-era Oracle SQL/database project with ER/schema materials, queries, sample data, and a technical report.

Five-stage pipelined RISC processor | Verilog, Icarus Verilog, GTKWave

Implemented and simulated a 32-bit five-stage Verilog processor; exercised arithmetic, load/store, and branch testbenches in Icarus Verilog.

RESEARCH MENTORING

Graduate Student Mentor, NSF Research Experiences for Undergraduates (REU) Program, Tennessee Technological University | Summers 2025 and 2026. Supported undergraduate research progress and technical development under faculty supervision.

CONFERENCE PRESENTATION

Poster presenter, "Investigating Adversarial Image Attacks in a Sensor Fusion Framework Using a Scaled Autonomous Vehicle Testbed," IFAC MECC 2025, Pittsburgh, Pennsylvania, October 5-8, 2025.

TECHNICAL SKILLS

Software and embedded systems: Python, C/C++, Linux, Git, ROS 2/SROS2 in QCar2 contexts, NVIDIA Jetson TX2/AGX Orin, Arduino/AVR.

Perception and evaluation: CSI and Intel RealSense cameras, 2D LiDAR, YOLOv5n/YOLOv8, OpenCV, PyTorch, scikit-learn, TensorFlow/Keras, NumPy/Pandas, study-specific TensorRT, CARLA, physical experiment design and analysis.

Digital design: Introductory Verilog, Icarus Verilog, GTKWave; Cadence VLSI coursework; MATLAB, LTspice.

INDUSTRY EXPERIENCE

Hardware Engineer Intern | Robi Axiata Limited, Dhaka, Bangladesh | December 2019-January 2020

- Prototyped AVR/Arduino systems and simulated hardware and communication behavior with MATLAB, LTspice, and Cisco Packet Tracer.

HONORS AND AWARDS

- Third runner-up, Banglalink SDG Hackathon: Code for a Cause, 2018.
- Board Scholarship, General Grade, Higher Secondary Certificate (HSC), Dhaka Board, 2016.
- Board Scholarship, General Grade, Secondary School Certificate (SSC), Dhaka Board, 2014.

SELECTED LEADERSHIP

- Treasurer, MIST Computer Club, 2020.
- Leading Organizer, National Collegiate Programming Contest (NCPC), 2020.
- Co-founder, Human Being Society Forever.